

# YONGJAE LEE

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 Berlin, Germany

## BIOGRAPHY

Yongjae Lee is a Ph.D. student in Computer Science at the Hasso Plattner Institute, University of Potsdam, Germany. He holds a M.Sc. in Industrial Data Science & Engineering and a B.Sc. in Industrial Engineering from Pusan National University, South Korea.

### Research Interests

- Business Process Management - BPM
- Process-Aware Agentic Systems
- Deep Learning for BPM

## EXPERIENCE

- **Data Systems, HPI**  2026.08 - Present  
*Scientific Assistant* Potsdam, Germany
  - Conducting research on Agentic Business Process Management, and Data Systems
- **Industrial Artificial Intelligence Research Institute**  2026.03 - 2026.06  
*Research Associate* Busan, South Korea
  - Collaborated with industry partners, including LG Electronics
  - Implemented deep learning models for industrial applications
- **Big Data Analytics Engineering Lab**  2022.09 - 2026.02  
*Researcher* Busan, South Korea
  - Conducted research on Business Process Management, and Deep Learning
  - Led 3-year research projects as a main researcher (budget, reports, and development)

## EDUCATION

- **Hasso Plattner Institute (University of Potsdam)** 2026.08 - Present  
*Ph.D. in Computer Science* Potsdam, Germany
  - Supervisor: [Prof. Dr. Matthias Weidlich](#)
- **Pusan National University** 2024.03 - 2026.02  
*M.Sc. in Industrial Data Science & Engineering* Busan, South Korea
  - Supervisor: [Prof. Dr. Hyerim Bae](#)
  - Thesis: *DHiM: A Dual-Modality Hierarchical Network for Multi-Perspective Business Process Anomaly Detection*
- **Pusan National University** 2018.03 - 2024.02  
*B.Sc. in Industrial Engineering* Busan, South Korea

## PROJECTS

- **Development of personalized smart operation logic based on customer product usage logs via process mining** 2025.03 - 2026.06  
*Funded by LG Electronics, Changwon*
  - Developed a personalized product control system using Process Mining and Multimodal AI techniques, aiming to enhance user experience.
- **A Study on the XPL (eXplainable Process Learning) based on Artificial Intelligence** 2023.03 - 2026.02  
*Funded by the National Research Foundation of Korea (NRF)*
  - Developed an explainable process learning methodology for process monitoring, detection, analysis, and improvement, based on artificial intelligence techniques.
- **Human Centered - Carbon Neutral Global Supply Chain Research Center** 2023.06 - 2025.02  
*Funded by the National Research Foundation of Korea (NRF)*
  - Engaged in a national research project to develop key technologies for human safety and environmental sustainability across integrated global maritime supply chains.

## PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PREPRINT

- [J.2] Park, H., Sim, S., Park, K., Lee, Y., and Park, E., and Bae, H. (2026, August) [Hybrid Temporal Autoencoder and Similarity Matching for Low Aggregation Level Long Time-Series Forecasting](#). *Journal of Forecasting*, 1–32.
- [P.2] Park, T., Lee, Y., Kim, D., and Bae, H. (2026, May). [LoopUS: Recasting Pretrained LLMs into Looped Latent Refinement Models](#). *arXiv preprint arXiv:2605.11011*.
- [C.4] Lee, Y., Park, E., Park, D., Kim, D., Choi, J., and Bae, H. (2026, May). [Process-Aware Procurement Lead Time Prediction for Shipyard Delay Mitigation](#). In *International Conference on Production Research* (pp. 3-13). Cham: Springer Nature Switzerland.
- [C.3] Park, T., Lee, Y. (co-first), Park, D., Kim, D., and Bae, H. (2025, December). [JustDense: Just using Dense instead of Sequence Mixer for Time Series analysis](#). In *2025 IEEE International Conference on Big Data (BigData)* (pp. 1172-1184). IEEE.
- [P.1] Lee, Y., Park, T. (co-first), Sim, S., and Bae, H. (2025, October). [Rewarding Structural Conformance of Reasoning using Process Mining](#). *arXiv preprint arXiv:2510.25065*.
- [C.2] Lee, Y., Kim, D., Kim, D., and Bae, H. (2025, August). [Multi-task Trained Graph Neural Network for Business Process Anomaly Detection with a Limited Number of Labeled Anomalies](#). In *International Conference on Business Process Management* (pp. 361-378). Cham: Springer Nature Switzerland.
- [J.1] Lee, Y., Park, K., Lee, H., Son, J., Kim, S., and Bae, H. (2024, December). [Identifying key factors influencing import container dwell time using eXplainable Artificial Intelligence](#). *Maritime Transport Research*, 7, 100116.
- [C.1] Iman, A. N., Kamal, I. M., Ramadhan, M. H., Kim, D., Lee, Y., and Bae, H. (2024, August). [Predictive process monitoring for remaining time prediction with transfer learning](#). *ICIC Exp. Lett.*, 18(8), 851-858.

## CONFERENCE PRESENTATIONS

- **Korea Institute of Industrial Engineers 2025 Fall Conference** 2025.11  
*DHiM: A Dual-Modality Hierarchical Network for Multi-Perspective Business Process Anomaly Detection* Gyeongju, South Korea
- **The 23rd International Conference on Business Process Management (BPM2025)** 2025.09  
*Multi-task Trained Graph Neural Network for Business Process Anomaly Detection with a Limited Number of Labeled Anomalies* Seville, Spain
- **The 28th International Conference on Production Research (ICPR28)** 2025.07  
*Process-Aware Procurement Lead Time Prediction for Shipyard Delay Mitigation* Bogota, Colombia
- **The 11th International Conference on Logistics and Maritime Systems (LOGMS2023)** 2023.09  
*Import Container Dwell Time: Analysis of Determinant Factors with Explainable Artificial Intelligence* Busan, South Korea

## PATENTS

- [1] Bae, H., Iman, A. N., Lee, Y., and Kim, D. Continual Learning Method and Device with Adaptive Memory Mechanism for Predictive Process Monitoring. *Korean Patent*, No.1020250026041.

## HONORS AND AWARDS

- **Grand Prize** August 2023  
*PNU Industrial AI Competition, Pusan National University*
- **Grand Prize** February 2023  
*Graduation Project, Dept. of Industrial Engineering, Pusan National University*

## TEACHING EXPERIENCE

- **Data Structures and Algorithms, Undergraduate Course** 2024.03 - 2024.06  
*Teaching Assistant, Pusan National University*
- **LG Electronics Process Mining Classroom Training (Supported by Celonis)** 2025.06 - 2025.07  
*Teaching Assistant, Pusan National University*

## SKILLS

- **Programming Languages:** Python, JavaScript
- **Deep Learning Frameworks:** PyTorch, PyTorch Geometric, TRL
- **Markup & Styling:** HTML/CSS, L<sup>A</sup>T<sub>E</sub>X, Markdown
- **Database:** MySQL, Neo4j
- **Other Tools & Technologies:** Plant Simulation (Siemens)

## LANGUAGES

Korean (Mother Tongue), English (Proficient), Chinese (Conversational)